

SYLLABUS :-

MEXXXX (New), Mechanics of Solids (3-1-0-4) Review of concept of stress and strain, Transformation of stress and strain: Transformation of stress and strain, principal stresses, principal strains, Mohr s circle for stress and strain. Torsion: Torsion of cylindrical bars, torsional stress, modulus of rigidity and deformation. Flexural loading: Shear and moment in beams; load, shear and moment relationship; flexure formula; shear stress in beams; differential equation of the elastic curve, deflection of beams. Combined loading: Axial and torsional; axial and bending; axial, torsional and bending. Column: Buckling of slender columns, Euler bucking load for different end conditions. Bending of Beams: Straight and asymmetrical bending, Shear center. Thick-walled cylinders. Energy methods: Castigliano s theorem Textbook: Mechanics of Materials: Beer, Johnston, DeWolf, Mazurek, Sanghi